

Android shim (VpnService + JNI)

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Revision: 2 Last modified: 2026-07-04T12:00:00Z **Rev 2:** Added §7.5 (battery-optimization / Doze / OEM-aggressive-killer mitigation — the foreground-service priority §7 already secures is necessary but not sufficient against OEM-specific background-kill behavior on MIUI/EMUI/One UI-class skins) and §14.1 (Google Play VPN-policy compliance requirements) — closing a gap identified in an independent enterprise-hardening pass over Volume 4.

Volume 4 (Clients) nano-detail specification — deepens the **Android TunnelPlatform shim** sketched in the pass-1 client overview [03 §5.2] into an implementation-ready design of the single platform-specific component that lets HelixVPN's shared Rust helix-core drive a system VPN on Android. SPEC ONLY: it describes **what to build** (Kotlin/Rust signatures, the channel contract, the VpnService lifecycle state machine, the JNI/UniFFI bridge, the tun fd handoff, protect() loop-avoidance, the foreground service, memory budget, error taxonomy, edge cases, and §11.4.169 test points) — it does **not** write the shipping product.

Boundary. This shim is the *only* Android-specific code in the client stack [03 §0, §4]. It **consumes:** the FFI surface + TunnelStatus enum owned by the core (helix-core/src/status.rs [01-ORCH §4.1]; the Dart-facing FFI [03 §3]), and the TunnelPlatform channel contract [03 §4]. It **owns:** the Kotlin VpnService + foreground service, the VpnService.Builder → ParcelFileDescriptor tun-fd handoff, the JNI/UniFFI bridge that hands that fd to helix-core and registers the protect() upcall, and the MethodChannel/EventChannel marshalling. It does **not** own WG crypto, the transport ladder, the orchestrator loops, or the kill-switch/DNS state machine — those are core [01-ORCH], identical on every platform.

Evidence base, cited inline by id. [03 §N] = final/03-client-core-and-ui.md (pass-1 client overview); [01-ORCH §N] = final/v02-data-plane/orchestrator-and-state.md (the TunnelStatus / orchestrator the shim drives); [01-TT §N] = final/v02-data-plane/transport-trait.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_UI §N] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [research-ios_android] = v09-research/research-ios_android.md (Apple DTS + Android VpnService primary-source research, all accessed 2026-06-25); [10-QA §N] = final/10-testing-acceptance-and-qa.md; [06-WBS §0.4] = final/06-phase0-spike-wbs.md (the §11.4.169 code map); [SYN §N] = cross-document synthesis. Anything not grounded in that base is tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

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0. Position, ownership, and invariants

The Android shim does **only three things** [03 §4]: (a) configure the OS tunnel (VpnService.Builder), (b) hand helix-core a packet fd + the protect() upcall, and (c) report lifecycle. Everything above it — ConnectButton, ExitPicker, settings, the TunnelStatus-driven UI — is shared Dart [03 §5, 04_UI §6]. The shim is the *only untyped seam* (native $\rightleftharpoons$ Dart) and is therefore covered by an on-device smoke test, never a mock-claiming-PASS (§11.4.27) [03 §4.1 O5].

0.1 Inherited invariants (the shim MUST keep)

#	Invariant	Source	Android-specific consequence
CI1	The shim does not own logic/status; it owns the OS tunnel lifecycle.	[03 §0.1]	The Kotlin VpnService creates the tun + foreground service; helix-core (the same .so as every platform) owns WG + transport + status.
O2	The shim hands the core a packet fd/pump; it never does crypto/obfuscation.	[03 §4.1]	coreAttachTun(fd); no WG/QUIC code in Kotlin.
CI2	The UI is a pure function of the core's status stream; never paint "connected" on intent.	[03 §0.1]	PlatformTunnelEvent.up ≠ TunnelStatus::Connected; Connected is emitted by the core <i>only</i> after the WG handshake reply [01-ORCH §5.4], never by the shim.
O1	startTunnel is idempotent + permission-aware: a denied VPN consent emits permissionDenied (not error) and leaves no half-open tun.	[03 §4.1]	§4, §9.1; cleanup on every exit path (§11.4.14).
O3	events() reports revoked when the OS kills the tunnel out-of-band.	[03 §4.1]	VpnService.onRevoke() → revoked (§4.4); never claim "connected".
O4	stopTunnel restores a quiescent OS state on every exit path.	[03 §4.1]	coreDetachTun() → close fd → stopForeground → remove notification (§4.5, §11.4.14).
I1	The transport never sees plaintext; the shim never inspects payload.	[01-TT I1]	The shim moves an opaque fd integer; it reads/writes no packet bytes (those flow inside the core).

0.2 What this document does NOT own

- The TunnelStatus enum + its Down.reason vocabulary — frozen in helix-core/src/status.rs [01-ORCH §4.1, §4.4]; the shim is a transparent carrier of those events to Dart.
 - The TransportConfig ladder, WG Tunn state machine, kill-switch/DNS state machine — core [01-ORCH §7, §8]; identical on every platform.
 - The Dart-side TunnelPlatform *caller* / Riverpod wiring [03 §8]; this doc owns the Android *implementation* the Dart contract dispatches to.
-

1. Component & process map

HelixVPN runs the Flutter engine, the Rust helix-core .so, **and** the VpnService in **one Android process** (the app's default process). This is a load-bearing decision (§16 D-ANDROID-1): a single process lets flutter_rust_bridge (frb) stream TunnelStatus and serve setShields/setExit **directly** from Dart to the in-process core [03 §3.2, §3.3], while the same in-process core holds the tun fd the VpnService produced — no cross-process IPC for status.

[MERMAID RENDER FAILED — block #1]

Two event paths leave the core (§8): **TunnelStatus** (core → frb → Dart, in-process, the truth about protection state), and **PlatformTunnelEvent** (Kotlin → EventChannel → Dart, the OS-tunnel lifecycle: up/down/ permissionDenied/revoked/error). They are distinct and MUST NOT be conflated (CI2).

2. The TunnelPlatform contract — Android impl surface

The frozen Dart contract [03 §4] is one MethodChannel (helixvpn/tunnel) + one EventChannel (helixvpn/tunnel/events). The Android side implements the method handler and the event sink.

2.1 Dart TunnelPlatform (consumed — frozen [03 §4])

```
// helix_core_ffi/lib/tunnel_platform.dart (frozen contract – restated, not redefined)
abstract class TunnelPlatform {
  Future<void> startTunnel(TunnelConfig
    cfg);      // OS permission + create TUN + link core + pump
  Future<void> stopTunnel();      // detach core,
    destroy TUN, release hold
```

```

    Stream<PlatformTunnelEvent> events(); // lifecycle
        only – up/down/permissionDenied/revoked/error
}

class TunnelConfig { // [03 §4]
    final String overlayIp; // e.g.
        100.64.0.7 (overlay /32, IPAM-assigned)
    final List<String> routes; // AllowedIPs
        into the tunnel (e.g. "10.10.0.0/24" | "0.0.0.0/0")
    final List<String> dnsServers; // tunnel DNS
        (anti-leak)
    final List<String> splitExcludeApps; // Android per-
        app bypass (package names)
    final int mtu; // 1280 MASQUE
        default, 1420 plain WG [03 §4, 01-TT §4.8]
    final String sessionOrMapToken; // handed
        straight to helix_core start (logic, not OS)
    // ... ctor as in [03 §4]
}

enum PlatformTunnelEventKind { up, down, permissionDenied,
    revoked, error }
class PlatformTunnelEvent { final PlatformTunnelEventKind kind;
    final String? detail; /* §11.4.6 honest reason */ }

```

2.2 The Android MethodChannel verb map

Dart call	MethodChannel method	Kotlin handler	Returns
startTunnel(cfg)	"startTunnel" (args = cfg map)	Plugin.onMethodCall → prepare check → start HelixVpnService (§4.2)	Future<void> (completes when the service is <i>started</i> , NOT when Connected)
stopTunnel()	"stopTunnel"	broadcast/ stopService → Service.onDestroy (§4.5)	Future<void>
events()	EventChannel onListen	Plugin registers an EventSink; the service posts PlatformTunnelEvents to it (§8.2)	Stream<PlatformTunnelEvent>

startTunnel's Future resolving means "the OS VPN bring-up was *requested* and consent is granted" — **not** that the tunnel is up. Up-ness arrives later as a PlatformTunnelEvent.up (carrier/tun ready) and, authoritatively, as a core TunnelStatus::Connected (WG handshake done) (CI2, §8).

2.3 Arg marshalling (the one untyped seam, made honest)

TunnelConfig crosses Dart→Kotlin as a Map<String, Any?> over the MethodChannel. The Kotlin side parses it into a typed TunnelConfig data class; a missing/invalid field is a **hard** parse error → PlatformTunnelEvent.error with an honest detail (§11.4.6 — never a silent default), never a half-open tunnel (O1). A round-trip fixture test (§15 UT) asserts every field survives the Map encoding byte-for-byte.

```
// HelixTunnelConfig.kt
data class HelixTunnelConfig(
    val overlayIp: String,
    val routes: List<CidrRoute>,           // parsed "a.b.c.d/n" →
        (InetAddress, prefix)
    val dnsServers: List<String>,
    val splitExcludeApps: List<String>,
    val mtu: Int,
    val sessionOrMapToken: String,
) {
    companion object {
        fun from(args: Map<String, Any?>): HelixTunnelConfig = /*
            typed parse; throw on missing/invalid */
    }
}
data class CidrRoute(val address: String, val prefix: Int)
```

3. The JNI / UniFFI bridge to helix-core

The shared Rust core is loaded as a native .so via JNI [03 §5 matrix, SYN §5, research-ios_android §3]. The **binding generator** is **UniFFI** for the typed Kotlin surface (decision D-CLIENT-3 [03 §14] — UniFFI gives typed Swift/Kotlin), with a **raw JNI export** reserved for the hot protect() upcall path where a per-socket UniFFI round-trip is undesirable (§6.3). frb generates the *Dart* surface [03 §3.2]; UniFFI generates the *native-shim* surface — two generators, one Rust source of truth [03 §3].

3.1 The exported Rust FFI surface used by the Android shim

This is the subset of the frozen FFI surface [03 §3.1] the Kotlin shim drives, plus the **one Android-specific addition** the contract needs (set_protect_callback, §6). TunnelStatus is the 5-variant core enum [01-ORCH §4.1] — kept consistent below (§3.3).

```

// helix-ffi/src/api.rs - UniFFI + frb generate from this
// (restated subset; full surface in [03 §3.1])
// All bodies live in helix-core; this crate is the EXPORTED
// surface only [03 §0].

pub struct ClientConfig { // [03 §3.1]
    pub map_path_or_session: String, // =
        TunnelConfig.sessionOrMapToken
    pub transport: String, // "auto" | "plain" |
        "masque" | ...
    pub mode: CoreMode, // Client (Android
        Access app) | Connector
}
#[derive(uniffi::Enum)] pub enum CoreMode { Client, Connector }

pub fn start(cfg: ClientConfig) -> Result<(), HelixError>; //
    logic bring-up (orchestrator start, [01-ORCH §2.1])
pub fn stop() -> Result<(), HelixError>; //
    graceful teardown

/// Shim handoff: the core NEVER opens the TUN; the VpnService
/// produces the fd and
/// hands it here. The core takes OWNERSHIP of the fd (closes it
/// on detach) – see §5.3.
pub fn attach_tun(fd: i32) -> Result<(), HelixError>; //
    [03 §3.1]
pub fn detach_tun() -> Result<(), HelixError>; //
    [03 §3.1]

/// Live status stream – maps to the orchestrator broadcast
/// channel [01-ORCH §4].
/// frb delivers this directly to Dart (in-process, §1); UniFFI
/// exposes it to Kotlin
/// too (the shim only needs it to drive PlatformTunnelEvent.up –
/// §8.3).
pub fn status_stream(sink: StreamSink<TunnelStatus>; //
    [03 §3.1]

/// ANDROID-SPECIFIC ADDITION (§6). The core invokes this callback
/// to protect every
/// outbound transport socket against the VPN routing loop BEFORE
/// it connects.
/// Registered once at attach time. No-op / returns Ok on non-
/// Android targets.
pub fn set_protect_callback(cb: Box<dyn ProtectCallback>; //
    NEW – Android contract extension
pub trait ProtectCallback: Send + Sync {
    /// Bind socket `fd` to the underlying physical network; true
    /// = protected.
    fn protect(&self, fd: i32) -> bool;
}

#[derive(uniffi::Error, Debug)] pub enum HelixError {
    Tun(String), KillSwitch(String), Dns(String),
    Internal(String) }

```

3.2 The UniFFI-generated Kotlin surface (primary path)

```
// generated by UniFFI from helix-ffi (do not hand-edit);
// HelixNative wraps it for ergonomics.
object HelixNative {
    init { System.loadLibrary("helix_core") } // load the .so
        once per process

    fun start(cfg: ClientConfig) // throws
        HelixException
    fun stop()
    fun attachTun(fd: Int) // hand the
        VpnService tun fd to the core (§5)
    fun detachTun()
    fun setProtectCallback(cb: ProtectCallback) // §6
    fun subscribeStatus(sink: TunnelStatusSink) // Kotlin-side
        status subscription (shim uses for .up only, §8.3)
}
interface ProtectCallback { fun protect(fd: Int):
    Boolean } // implemented by HelixVpnService.protect
    (§6.2)
```

3.3 TunnelStatus — consistency with the core (frozen)

The shim is a **transparent carrier** of the core's TunnelStatus; it never defines its own. The canonical enum is helix-core/src/status.rs [01-ORCH §4.1]:

```
// helix-core/src/status.rs (frozen, mirrored across the FFI [01-
// ORCH §4.1]) – restated for the binding
#[derive(Clone, Debug, PartialEq, Eq)]
pub enum TunnelStatus {
    Connecting, // dialling a
        rung (carrier handshake in flight)
    Handshaking, // carrier up;
        WG Noise IK handshake in flight
    Connected { transport: String, rtt_ms: u32 }, // e.g.
        ("masque-h3", 23) – WG established
    Reconnecting, // a working
        tunnel dropped; re-dialling (ladder/backoff)
    Down { reason: String }, // terminal for
        this attempt; stable-prefix reason [01-ORCH §4.4]
}
```

Consistency note (§11.4.6). The pass-1 client overview presents an *extended* Dart mirror that adds Disconnected, a path field on Connected, and a Danger{kind} variant for the UI palette [03 §3.1 status-enum note]. Those are **FFI-presentation extensions** layered on top of this 5-variant core enum; this shim spec anchors on the **core 5-variant TunnelStatus** [01-ORCH §4.1] as the source of truth, exactly as the orchestrator emits it. The shim forwards whatever

variant the core emits, unmodified — if/when the extended variants are ratified into the core enum, the shim needs **no change** (it is a transparent carrier). The Android shim emits its own **separate** PlatformTunnelEvent for OS-tunnel lifecycle (§8.2); it never synthesises a TunnelStatus.

3.4 Raw-JNI fallback (the hot protect() path)

For the per-socket protect() upcall (§6) the core may, instead of a UniFFI callback object, call a #[no_mangle] extern "system" JNI function or — as a perf option — the native protectFromVpn symbol (§6.3). The raw-JNI export shape:

```
// helix-ffi/src/jni_android.rs (cfg(target_os="android")) –  
    fallback / hot-path only  
#[no_mangle]  
pub extern "system" fn  
    Java_dev_helixvpn_core_HelixNative_coreAttachTun(  
        env: JNIEnv, _class: JClass, fd: jint) -> jint { /*  
        attach_tun(fd) → 0 | errcode */ }
```

UniFFI is the **primary** typed surface; the raw JNI export exists only where the hot path or a UniFFI gap (UNVERIFIED: UniFFI callback-interface overhead per upcall — measure in BENCH, §15) justifies it.

4. VpnService lifecycle state machine

4.1 States (Kotlin-side, distinct from TunnelStatus)

The VpnService has its **own** lifecycle — the OS-tunnel layer — that the shim projects to PlatformTunnelEvent. It is *not* TunnelStatus: the tun can be Up (fd established, core attached) while TunnelStatus is still Handshaking (WG not yet established). The UI trusts TunnelStatus for protection state (CI2); the shim's lifecycle drives only the foreground service + the up/down/revoked events.

[MERMAID RENDER FAILED — block #2]

4.2 startTunnel → onStartCommand (the bring-up)

```
// HelixVpnService.kt  
class HelixVpnService : VpnService(), HelixNative.ProtectCallback  
{  
  
    override fun onStartCommand(intent: Intent?, flags: Int,  
        startId: Int): Int {  
        // null intent ⇒ START_STICKY redelivery after process death  
        (§9.3) – restore or stop.
```

```

    val cfg = HelixTunnelConfig.from(intent?.bundleExtras() ?:
        return restoreOrStop())

    // 1. Foreground FIRST (must call startForeground within the
    // OS window, §7) – survive bg-kill.
    startForeground(NOTIF_ID,
        buildPersistentNotification()           // §7

    // 2. Build + establish the TUN (§5). Null ⇒ another VPN
    // active / invalid config (§12 E3).
    val pfd: ParcelFileDescriptor = buildTun(cfg)
    ?: return failAndStop("establish() returned
        null")           // ErrorState

    // 3. Register protect upcall BEFORE the core opens any
    // transport socket (§6).
    HelixNative.setProtectCallback(this)

    // 4. Hand the fd to the core; take ownership via detachFd
    // (§5.3). Then start the core logic.
    val fd =
        pfd.detachFd()                               //
        native now owns close()
    HelixNative.start(cfg.toClientConfig())           //
        orchestrator start [01-ORCH §2.1]
    HelixNative.attachTun(fd)                         //
        core pumps fd ⇌ WG ⇌ transport

    emitEvent(PlatformTunnelEventKind.up,
        null)           // OS-tunnel up (NOT
        "connected", CI2)
    return
        START_STICKY                               //
        §9.3 (with redelivery handling)
}
}

```

Ordering is load-bearing: **foreground first** (or the OS may kill a just-started background service before `establish()`), **establish before attach** (the core needs a live fd), **protect-callback before start** (the core may open a transport socket the instant it starts — it must be protectable, §6).

4.3 Handshaking → Connected is the core's call, not the shim's

The shim emits `PlatformTunnelEvent.up` once the *carrier* (fd + core attach) is ready. The user-visible "Connected" (green) is driven by the core's `TunnelStatus::Connected` over frb [01-ORCH §5.4, §3 §3.2] — emitted only after the **WG handshake reply** lands, never on tun-up. This split is the exact mechanism that prevents "UI says connected while the tunnel is actually down" (CI2) [03 §3.3].

4.4 onRevoke (O3)

```
override fun onRevoke()
{
    // OS or another VPN app took over, or
    // user revoked consent
    HelixNative.detachTun(); HelixNative.stop()
    emitEvent(PlatformTunnelEventKind.revoked, "os_revoked")
    stopForeground(STOP_FOREGROUND_REMOVE); stopSelf()
}
```

onRevoke is the out-of-band kill path (O3). It also covers the control-plane revocation reflected as device.revoked (doc 02) when that propagates to a tunnel teardown — though a *cryptographic* WG revoke surfaces first as the core's Down{reason:"auth-failed"} over TunnelStatus [01-ORCH §5.4]; onRevoke is the *OS-level* VPN-permission revoke. Both end in a cleaned-up quiescent state (O4).

4.5 stopTunnel / onDestroy (O4)

```
override fun onDestroy() { // user stop OR system
    // teardown

    HelixNative.detachTun() // core closes the
    // owned fd (§5.3)

    HelixNative.stop() // orchestrator
    // graceful stop [01-ORCH §2.4]

    stopForeground(STOP_FOREGROUND_REMOVE) // remove the
    // persistent notification (§7)
    emitEvent(PlatformTunnelEventKind.down, null)
    super.onDestroy()
}
```

Every exit path (stopTunnel, onRevoke, onDestroy, ErrorState, process-death) **MUST** leave: **no orphan fd** (core closed it), **no foreground notification, no leaked routes/DNS** (those were tun-scoped and die with the tun). A post-stop sanity assertion (§15 SEC/SC) FAILS the test if any orphan survives (§11.4.14).

5. VpnService.Builder → tun fd handoff

5.1 Building the interface

VpnService.Builder configures a Layer-3 interface: at least one address + routes/MTU before establish(), which returns a ParcelFileDescriptor for the TUN [research-ios_android §3, 04_UI §6 row]. The app reads/writes **raw IP packets** on that fd — there is no per-socket model [research-ios_android §3].

```

private fun buildTun(cfg: HelixTunnelConfig):
    ParcelFileDescriptor? {
    val b = Builder()
        .setSession("HelixVPN") //
            label in the system VPN UI
        .addAddress(cfg.overlayIp, 32) //
            overlay /32 (IPAM-assigned)
        .setMtu(cfg.mtu) //
            1280 MASQUE / 1420 plain [01-TT §4.8]
        .setBlocking(false) //
            non-blocking fd (core pumps async)
    cfg.routes.forEach { b.addRoute(it.address, it.prefix) } //
        AllowedIPs (e.g. 0.0.0.0/0 full-tunnel)
    cfg.dnsServers.forEach { b.addDnsServer(it) } //
        tunnel DNS (anti-leak, §10.2)
    applySplitTunnel(b, cfg.splitExcludeApps) //
        §10.1 (per-app bypass)
    // Allow the app's OWN sockets to bypass the tun so the
    // transport can reach the gateway:
    // handled by protect() per-socket (§6), NOT by a bypass route –
    // keeps the kill-switch tight.
    return b.establish() //
        null ⇒ no VpnService permission / invalid
}

```

5.2 The fd is Layer-3 IP, owned by the core

Once attached, helix-core does read(2)/write(2) of IP packets directly on the fd inside its loop A/B [01-ORCH §3.1, §3.2] — exactly the tun2socks-style native consumption pattern [research-ios_android §3]. The Kotlin side never touches packet bytes (I1, O2).

5.3 fd ownership: detachFd() (recommended) vs getFd()

Option	Ownership	Who closes	Trade-off
detachFd() (recommended)	transfers the raw int fd to native; the ParcelFileDescriptor is invalidated	the core closes it in detach_tun()	clean single-owner lifecycle; the core controls fd close exactly when it stops pumping [research-ios_android §3]

Option	Ownership	Who closes	Trade-off
getFd()	Kotlin keeps the ParcelFileDescriptor alive	Kotlin must keep pfd referenced and close() it on stop	risks a double-close or a leak if the service dies before Kotlin closes; needs careful lifecycle bookkeeping

Decision (D-ANDROID-2, §16): use `detachFd()` → the core owns and closes the fd via `detach_tun()`. The Kotlin side keeps **no** `ParcelFileDescriptor` reference after `detachFd()` (it is invalidated). Edge: if the core process dies, the fd is reclaimed at process death — acceptable, no cross-process leak.

[MERMAID RENDER FAILED — block #3]

6. protect() — the loopback-avoidance contract

6.1 Why it is mandatory

`VpnService.protect(socket)` binds a socket to the **underlying physical network**, so the tunnel's own outbound transport packets (to the gateway) do **not** get routed back into the tun — which would infinite-loop. It **MUST** be called **before** the socket connects/sends, for **every** outbound transport socket the core opens [research-ios_android §3]. Without it the tunnel cannot carry its own carrier traffic. This is the single Android-specific obligation that forces a **core → shim upcall** (the core opens the socket inside `helix-transport::dial()` [01-TT §6], so only the core knows the fd; only the shim can `protect()` it).

6.2 The upcall (primary path: `VpnService.protect`)

The core invokes the registered `ProtectCallback` synchronously, before `connect()`, for each transport socket:

```
// HelixVpnService implements HelixNative.ProtectCallback
override fun protect(fd: Int): Boolean = protect(fd) //
    VpnService.protect(int) - true on success
```

```
// inside helix-transport dial(), cfg(target_os="android"):
    protect every socket before connect
let sock = socket2::Socket::new(domain, ty, proto)?;
if !PROTECT.get().map_or(false, |cb|
    cb.protect(sock.as_raw_fd())) {
    return Err(TransportError::Io(io::Error::other("protect()
        failed"))); // never connect an unprotected socket
}
sock.connect(&peer.into())?; // [01-TT §6.1]
```

A failed `protect()` is a **hard** dial error (`TransportError::Io` → ladder/backoff [01-TT §7, 01-ORCH §7]) — the core **MUST NOT** connect an unprotected socket (it would leak onto the underlay and loop). This is verified by the §15 SEC test (a captured pcap shows transport packets egress the physical iface, not the tun).

6.3 Hot-path option: native `protectFromVpn` (perf)

`VpnService.protect()` ultimately calls the C function `protectFromVpn` (`NetdClient.h`); a native core can bind that symbol and protect sockets entirely in native space, avoiding a JNI/UniFFI upcall on the hot connect path [research-ios_android §3]. **Decision (D-ANDROID-3, §16)**: use the `ProtectCallback` upcall as the **safe default** (`protect` is not on the per-packet hot path — it fires once per *socket* / per dial, which is rare); adopt native `protectFromVpn` only if BENCH (§15) shows the upcall cost matters. UNVERIFIED: `protectFromVpn`'s symbol stability across Android versions / OEM images — it is **not** a public NDK symbol; binding it is a documented community pattern, not a guaranteed-stable ABI [research-ios_android §3]. Treat as an optimization behind a measured gate, never the default (§11.4.6).

6.4 Re-protect on roam

When the device's primary network changes (Wi-Fi→cellular), the previously-protected socket's underlying network may vanish → the carrier dies → the core re-dials [01-ORCH §7.5] → a **new** transport socket is opened → `protect()` is invoked again on the **new** network. The shim feeds the roam signal to the core via a `ConnectivityManager.NetworkCallback` → `RoamDetected{new_local}` [01-ORCH §7.5, §9]; the core's < 3 s roam recovery [01-ORCH §7.1] then re-establishes a freshly protected socket. No special shim logic beyond delivering the roam event and answering the `protect` upcall.

7. Foreground service — surviving background-kill

A VPN tunnel must keep running when the app is backgrounded; Android only keeps a service alive in the background if it is a **foreground service** with a persistent notification [research-ios_android §3 implies via VpnService foreground practice; 03 §5.2 startForeground].

```
private fun buildPersistentNotification(): Notification {
    // Notification channel created once at app start (API 26+).
    return NotificationCompat.Builder(this, CHANNEL_ID)
        .setSmallIcon(R.drawable.ic_helix_shield)
        .setContentTitle("HelixVPN")
        .setContentText("Protected") // updated from
            TunnelStatus over a status subscription (§8.3)
        .setOngoing(true)
        .setCategory(NotificationCompat.CATEGORY_SERVICE)
        .build()
}
// onStartCommand: startForeground(NOTIF_ID,
    buildPersistentNotification() [, foregroundServiceType])
```

Concern	Requirement	Note
Foreground window	startForeground() MUST be called promptly inside onStartCommand (before establish() in §4.2)	else the OS may kill the just-started service
Notification channel	created once (API 26+) before first startForeground	POST_NOTIFICATIONS runtime permission on API 33+
Foreground service type	a foregroundServiceType MUST be declared (manifest) + passed on API 34+	UNVERIFIED: the exact constant for a VPN foreground service on API 34+ — verify against the latest Android docs (§11.4.99) before coding; declare the matching FOREGROUND_SERVICE_* permission. Do not hardcode a guessed type.
Restart	START_STICKY so the OS recreates the service after a low-memory kill (§9.3)	with null-intent redelivery handling
Notification text	reflects TunnelStatus (Connecting/	the shim keeps a lightweight Kotlin-side

Concern	Requirement	Note
	Connected/ Reconnecting)	status subscription only to update this text (§8.3)

Android memory posture (contrast iOS). Android VpnService runs as a normal foreground service — there is **no** iOS-NE-style hard ~15/50 MiB jetsam cap; the standard Android per-app memory / lowmemorykiller applies, which is far more generous, and a foreground service is high-priority [research-ios_android §3]. The severe memory constraint is **iOS-only**; on Android the foreground-service priority is the survival mechanism, not a tight RSS budget (§11). The shim still bounds buffer pools in the core (size receive buffers to MTU+headroom [research-ios_android §3, §4]) because the **same lean Rust core** ships on both.

7.5 Battery-optimization / Doze / OEM-aggressive-killer mitigation

§7's foreground-service + persistent-notification discipline is AOSP's own background-kill defense and is **necessary but not sufficient**: several major OEM Android skins (commonly cited: Xiaomi MIUI, Huawei EMUI/HarmonyOS-for-phones, Samsung One UI, OPPO ColorOS, Vivo FuntouchOS) ship an **additional, non-AOSP battery-management layer** that can kill or freeze a foreground service — including a VPN foreground service — despite AOSP Doze/App-Standby already exempting active VPN sessions. This is a well-known, real-world class of complaint for VPN apps specifically (a foreground "Protected" notification stays visible while the tunnel is silently frozen by the OEM layer) and **MUST** be mitigated, not merely documented as a known limitation:

Mitigation	Mechanism	Note
Request battery-optimization exemption	Intent(Settings.ACTION_REQUEST_IGNORE_BATTERY_OPTIMIZATIONS) (AOSP-standard, requires the REQUEST_IGNORE_BATTERY_OPTIMIZATIONS manifest permission — a Google Play policy-sensitive permission, §14.1)	show during onboarding with honest explanation of why general "improvement" performance dialocons honest user decline func

Mitigation	Mechanism	Note
Detect OEM skin + surface a targeted deep link	Build.MANUFACTURER/Build.BRAND sniff (closed, maintained list) → deep-link to the OEM's own "autostart"/"protected apps"/"battery saver whitelist" settings screen (package+activity names are OEM- and OS-version-specific, UNVERIFIED — maintain a small, tested per-OEM table, degrade to the generic Android battery-settings screen if the OEM-specific intent is not resolvable)	is wa may kille this actu MIU class — th exen along not c the O sepa auto prot apps hone evid base class (\$11 neve gues dres fact; class is a l on a obse state trans tagg such
Detect a suspected OEM-kill post-hoc	if the shim transitions Up → Restarting (§9.3) with no corresponding user action AND no onRevoke/network-loss signal, classify as suspected_oem_kill rather than a generic restart, and surface a one-time in-app notice pointing the user back to the OEM settings deep link	
Reconnect after an OEM-kill	on Restarting with a persisted last-good TunnelConfig (§9.3), reconnect automatically if the user's last intent was "connected" — same rule as D-ANDROID-4, so an OEM-kill and a plain process-death recover identically	§9.3

This closes the operational reality that "foreground service" is a *necessary* survival mechanism (§7) but the *sufficient* one on OEM-modified Android requires an explicit exemption request + a per-OEM escape hatch; shipping only the AOSP-standard foreground service and calling the background-kill risk "handled" would be an enterprise-hardening gap for the very OEM skins that are common in several of HelixVPN's target markets (self-hosted/home-lab operators on consumer Android devices).

8. Status & event flow (the two channels)

8.1 The two distinct streams

Stream	Producer → consumer	Transport	Carries	Truth about
TunnelStatus	helix-core → Dart	frb StreamSink, in-process [03 §3.2]	Connecting/Handshaking/ Connected{transport,rtt}/ Reconnecting/ Down{reason} [01-ORCH §4.1]	protected state (CI2) the UI paints this (CI2)
PlatformTunnelEvent	Kotlin shim → Dart	EventChannel helixvpn/ tunnel/ events [03 §4]	up/down/permissionDenied/ revoked/error	OS-tunnel lifecycle (consent fd, foreground service)

[MERMAID RENDER FAILED — block #4]

8.2 PlatformTunnelEvent emission rules

The shim emits exactly one event per OS-tunnel transition (§4): up on establish()+attach success; down on graceful stop; permissionDenied on declined consent; revoked on onRevoke; error{detail} on establish()==null / attach failure / config-parse failure. detail is an **honest** string (§11.4.6 — never "likely"/"maybe"). Events post to the registered EventSink on the platform thread; if no listener is attached yet, the latest lifecycle event is buffered (UNVERIFIED: buffer depth = 1 latest, sufficient because the UI re-reads latest on listen — calibrate in §15 UT).

8.3 Why the shim also subscribes to TunnelStatus (notification text)

The shim keeps a **lightweight** Kotlin-side TunnelStatus subscription (HelixNative.subscribeStatus) for **one** purpose: updating the persistent notification text (Connecting.../Protected...masque-h3 · 23 ms/Reconnecting...). This is presentation only; the shim never *re-derives* protection state nor feeds it back to the UI (the UI already has the authoritative frb stream, CI2). A RecvError::Lagged on this subscription is treated as "re-read latest", never an error [01-ORCH §4.6].

9. Permission, always-on, and boot

9.1 Permission flow (VpnService.prepare)

VpnService.prepare(context) returns an Intent if the user has not yet consented to this app acting as a VPN (must be launched via startActivityForResult), or null if already consented. Declined consent → PlatformTunnelEvent.permissionDenied, **no** service start, no half-open tun (O1).

```
fun startTunnel(cfg): Future {  
    val prep = VpnService.prepare(context)           // null =  
                                                already granted  
    if (prep != null) { launchConsent(prepare) /* on result:  
                                                granted→continue | denied→permissionDenied */ }  
    else startService(cfg)  
}
```

9.2 Always-on VPN & lockdown (the Android kill-switch)

Android's system **Always-on VPN + Block connections without VPN (lockdown)** setting is the OS-level kill-switch on Android [04_ARCH §5.6 "kill-switch+DNS driven by core state machine"; the OS lockdown is the platform mechanism]. When lockdown is on, the OS starts HelixVpnService via an OS-issued intent (no UI) and drops all non-VPN traffic. The shim MUST therefore:

- handle a service start with **no** UI in the foreground (start foreground + build the tun from a **persisted** last-good TunnelConfig, §9.3);
- never assume the Flutter UI is alive when onStartCommand fires.

The core's own fail-closed kill-switch posture [01-ORCH §8] composes with the OS lockdown: the core drops on Reconnecting/Down regardless, and Android lockdown additionally blocks at the OS routing layer. The two are complementary, not redundant.

9.3 Process death & START_STICKY

If the OS kills the service under memory pressure, START_STICKY recreates it and re-delivers onStartCommand with a **null** intent. The shim must then:

[MERMAID RENDER FAILED — block #5]

The last-good TunnelConfig is persisted (the sessionOrMapToken is a secret → stored in EncryptedSharedPreferences / Keystore-backed store, never plaintext, §11.4.10) so always-on/lockdown and post-kill restarts can re-establish without the UI. UNVERIFIED: whether to

auto-reconnect on every redelivery vs only under lockdown — surface as D-ANDROID-4 (§16); default = reconnect only if the user's last intent was "connected" (mirrors [03 §8.4] intent honesty).

10. Split-tunnel, DNS, routes, MTU

10.1 Per-app split tunnel

VpnService.Builder supports **either** `addDisallowedApplication(pkg)` (exclude-list: everything tunnels except these) **or** `addAllowedApplication(pkg)` (include-list: only these tunnel) — the two modes are **mutually exclusive** (you cannot mix them on one Builder).

```
private fun applySplitTunnel(b: Builder, exclude: List<String>) {  
    // HelixVPN MVP uses the exclude model (cfg.splitExcludeApps)  
    [03 §4 TunnelConfig.splitExcludeApps].  
    exclude.forEach { pkg -> runCatching {  
        b.addDisallowedApplication(pkg) } } // ignore unknown pkg  
    (honest skip)  
}
```

A package that is not installed throws `NameNotFoundException`; the shim skips it with an honest log, never aborts the whole tunnel (§11.4.6 — but the skip is *recorded*, not silent). Include-mode (`addAllowedApplication`) is a Phase-2 config knob (D-ANDROID-5, §16).

10.2 DNS leak protection

Tunnel DNS is set via `addDnsServer(it)` for each `cfg.dnsServers`. Because all routes (including `0.0.0.0/0` in full-tunnel mode) go into the tun and no bypass route is added, off-tunnel `:53` cannot egress — this is the Android realisation of the core's "force tunnel DNS; block plaintext :53 off-tunnel" guarantee [04_ARCH §5.6, 01-ORCH §8.3]. Split-tunnel excluded apps deliberately bypass (per user config); that is an explicit user choice, surfaced honestly in the UI, not a leak.

10.3 Routes & MTU

- Routes = `cfg.routes` (AllowedIPs): `0.0.0.0/0` (+ `::/0` for v6) for full-tunnel privacy-exit mode; specific overlay CIDRs (e.g. `10.10.0.0/24`) for joined-network-only mode [SYN §1].
- MTU = `cfg.mtu`: 1280 over MASQUE, 1420 plain WG [01-TT §4.8] — the value the core computed from `effective_mtu()` of

the active transport and pushed into the TunnelConfig [01-ORCH §6.6]. The shim does not invent the MTU.

11. Memory & performance budget

Budget	Android target	Basis
Core steady-state RSS	low-tens MB; no hard jetsam cap like iOS NE	Android VpnService is a normal foreground service; lowmemorykiller is generous [research-ios_android §3]
Buffer pools	bounded; receive buffer sized to MTU+headroom	same lean Rust core as iOS; bound pools to avoid spike-kills [research-ios_android §3, §4]
Foreground priority	high (foreground service) protects against bg-kill	§7
App install size	shares the --split-per-abi lean Rust staticlib	[03 §10]
Packet path	core does read/write on the fd in its own tokio threads	no Kotlin on the per-packet path (§5.2)
protect() cost	once per transport socket (rare), not per packet	§6.3 (native protectFromVpn only if BENCH demands)

The binding memory constraint of the program is **iOS NE**, not Android [research-ios_android §3, §4; CI4 03 §0.1]. On Android the shim's job is survival (foreground service) and correctness (protect, fd lifecycle), not a tight RSS ceiling. The Android MEM test (§15) is a **regression guard** (dumpsys meminfo, no leak over a soak), not a make-or-break gate (the make-or-break G3 is iOS-only [03 §5.1]).

12. Error taxonomy & edge cases

#	Condition	Required behaviour	Event / state	Source
E1	VPN consent declined	no service start; no half-open tun (O1)	permissionDenied → Idle	§9.1

#	Condition	Required behaviour	Event / state	Source
E2	Config map missing/invalid field	hard parse error; no tun	error{detail} → Idle	§2.3
E3	establish() returns null (another VPN active / invalid Builder)	cleanup; no orphan fd	error{"establish_null"} → Idle	§5.1, [research-ios_android §3]
E4	protect() fails on a transport socket	core does NOT connect; ladder/backoff	core TransportError::Io → Reconnecting	§6.2
E5	onRevoke (OS / another VPN / user revoke)	coreStop; quiescent restore (O3,O4)	revoked → Idle	§4.4
E6	Roam (Wi-Fi→cellular) while connected	feed RoamDetected; core re-dials + re-protects; < 3 s recovery	core Reconnecting→Connected	§6.4, [01-ORCH §7.5]
E7	Process death (lowmemorykiller) while connected	START_STICKY redeliver; restore from persisted cfg OR stop cleanly	Restarting → Establishing/Stopped	§9.3
E8	Always-on/ lockdown start with no UI	start foreground + build tun from persisted cfg without the Flutter engine	Establishing (headless)	§9.2
E9	WG cryptographic revoke (device.revoked)	core emits Down{auth-failed} over TunnelStatus; kill-switch stays closed	TunnelStatus::Down + shim down	[01-ORCH §5.4, §8.5]
E10	stopTunnel while mid-handshake	detach+stop cleanly;	down → Idle	§4.5

#	Condition	Required behaviour	Event / state	Source
E11	Unknown split-exclude package	close fd; no orphan skip with honest record; tunnel proceeds foreground still required;	(logged)	§10.1
E12	Notification permission denied (API 33+)	surface the limitation honestly; tunnel proceeds if OS allows	(logged)	§7 (UNVERIFIED: per-OEM behaviour)

Error-classification rule (§11.4.6): the shim never labels a failure "transient/flaky"; error.detail is a concrete observed cause or an explicit UNKNOWN:/PENDING_FORENSICS: marker with a tracked follow-up.

13. Threading & cancel model

- onStartCommand runs on the **main thread**; establish() is cheap; the per-packet pump runs inside the **core's own tokio threads** after attach_tun (the shim never blocks the main thread with packet I/O) [01-ORCH §3].
- The frb TunnelStatus stream is delivered off the platform thread by frb [03 §3.2]; the Kotlin-side notification subscription (§8.3) hops to the main thread only to update the notification.
- protect() upcalls are synchronous and brief (one protect(int) syscall); they occur on whatever core thread opened the socket — the ProtectCallback impl MUST be thread-safe and reentrancy-free (it only calls VpnService.protect, which is thread-safe).
- Cancellation: stopTunnel/onDestroy/onRevoke call detach_tun() +stop(), which cancel the core's CancellationToken and join its loops within shutdown_grace (2 s default) [01-ORCH §2.4]; the fd is closed by the core on detach (§5.3). No core thread outlives the service.

14. Manifest & build requirements

```
<!-- AndroidManifest.xml (essentials) -->
<uses-permission
    android:name="android.permission.FOREGROUND_SERVICE"/>
<!-- API 34+: declare the matching FOREGROUND_SERVICE_<type>
    permission (UNVERIFIED exact type, §7) -->
<uses-permission
    android:name="android.permission.POST_NOTIFICATIONS"/>
<!-- API 33+ -->

<service
    android:name=".HelixVpnService"
    android:permission="android.permission.BIND_VPN_SERVICE"
    android:exported="false">
    <!-- android:foregroundServiceType="..." on API 34+
        (UNVERIFIED exact value, §7) -->
    <intent-filter><action android:name="android.net.VpnService"/>
    </intent-filter>
</service>
```

- The .so is the shared lean Rust helix-core staticlib→cdylib for Android ABIs (arm64-v8a, armeabi-v7a, x86_64), --split-per-abi to keep install size lean [03 §10]; same opt-level="z"/"s" + LTO + panic=abort + strip profile as iOS for code-size/footprint parity [research-ios_android §2].
- UniFFI Kotlin bindings + frb Dart bindings are **generated and drift-checked** in the local melos build (no active CI per §11.4.156 [SYN §9]); a binding drift vs the Rust helix-ffi source is a build-blocking finding [03 §11].

14.1 Google Play VPN-policy compliance (engineering-relevant subset)

Google Play applies a dedicated review policy to any app declaring BIND_VPN_SERVICE — this is a build/ship dependency the WBS must schedule for, not a detail discovered at submission time:

Requirement	What it means for this shim
Prominent disclosure + in-app disclosure of VPN behavior	the onboarding flow (Volume 10 v10-design/ux-flows-and-interaction.md) must state what the app does before first connect — consistent with the no-logging-as-code privacy stance this project already leads with, so disclosure is a

Requirement	What it means for this shim
Data-safety form (Play Console) accuracy	documentation task, not a design conflict must match the ACTUAL data handled — no durable connection/traffic log (per v05-security/no-logging-as-code.md) is a claim the Play Store data-safety form must reflect precisely; a mismatch between the form and shipped behavior is a policy violation independent of code correctness
REQUEST_IGNORE_BATTERY_OPTIMIZATIONS permission scrutiny	Play policy restricts this permission to apps with a legitimate need; a VPN foreground service (§7.5) is an accepted use case, but the permission request flow must present the honest justification dialog (§7.5) rather than a generic one, since Play reviewers evaluate the user-facing justification text
Foreground service type declaration (API 34+)	the manifest <code>foregroundServiceType</code> (§14, UNVERIFIED exact constant) must be one Play accepts for a VPN class service — verify against the current Play Console policy + Android docs together (§11.4.99), not Android docs alone, since Play policy can be stricter than the platform API surface
Repeated policy-violation risk	a VPN app that is later found to log traffic content despite claiming not to is a Play <i>policy</i> strike, not merely a bug — this is additional motivation (beyond the

Requirement	What it means for this shim
	architectural no-logging-as-code mandate) for the CI schema-lint in v05-security/no-logging-as-code.md to stay green at every release

This does not change any technical design in this document; it is recorded here so 06-phase0-spike-wbs.md/07-phase1-mvp-wbs.md schedule the Play Console data-safety-form + policy-review lead time as a real dependency of the Phase-1 MVP ship date.

15. Test points (§11.4.169)

§11.4.169 (a LIVE constitution anchor) mandates that every workable item declare its required test types from the closed vocabulary [06-WBS §0.4, 10-QA §2]; the only permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason, never a silent gap. The shim is the **only untyped seam**, so its evidence bar is the on-device smoke (O5). Four-layer enforcement per §11.4.4(b) applies to every closure.

Code	Type	Android-shim test point	Evidence (§11.4.5/.69/.107)
UT	unit	HelixTunnelConfig.from(map) round-trip (every field survives, §2.3); PlatformTunnelEvent mapping; split-tunnel mode-exclusivity guard; detail honesty (no forbidden vocab)	gradle test / Robolectric logs
INT	integration (real system, infra via containers submodule §11.4.76)	the Kotlin shim + real helix-core .so on an emulator: startTunnel → establish() → attachTun → core reaches a real edge booted on-demand	logcat + netns/ edge round-trip
E2E	end-to-end	on-device: real Access flavor → connect → curl/app traffic reaches an authorized LAN host through the tun; plain-UDP blocked path escalates to MASQUE [01-ORCH §7]	window-scoped MP4 (§11.4.159) + capture

Code	Type	Android-shim test point	Evidence (§11.4.5/.69/.107)
FA	full-automation (§11.4.25/.52/.98, deterministic §11.4.50)	scripted (uiautomator/adb) connect→status→disconnect, N=3 identical PlatformTunnelEvent + TunnelStatus traces, zero human-in-loop	3× identical trace artifacts
SEC	security (§11.4.10 + security submodule)	(a) pcap proves transport sockets egress the physical iface (protect works, §6) not the tun; (b) no off-tunnel :53 leak (§10.2); (c) sessionOrMapToken never in logs / plaintext prefs (§9.3, §11.4.10)	pcap + secret-leak grep-empty
STRESS	stress (§11.4.85)	sustained connect/disconnect churn (≥ 100 cycles); fd-leak census (/proc/<pid>/fd count stable); bounded RSS	fd-count + RSS CSV
CHAOS	chaos (§11.4.85)	iface flap (toggle Wi-Fi/cellular) mid-transfer → < 3 s recovery (§6.4, E6); process-kill (am kill) → START_STICKY restore (E7); onRevoke mid-transfer (E5)	recovery trace + logcat
CONC	concurrency / atomicity	concurrent startTunnel/stopTunnel calls (idempotency, O1); protect() upcalls from multiple core threads thread-safe	harness log
RACE	race / deadlock	no deadlock between onDestroy (detach) and an in-flight protect() upcall; fd close-vs-use ordering	stress harness + tsan-style assertions
MEM	memory	regression guard: dumsys meminfo RSS stable over a soak; no fd/handle leak; not a make-or-break (iOS owns G3)	dumsys meminfo series
BENCH/ PERF	benchmarking / performance	through-tunnel throughput on device; protect() upcall cost (decides D-ANDROID-3 §6.3); attach→Connected latency	throughput/latency CSV
UI/UX	UI / UX	the shared Flutter ConnectButton/StatusChip	golden + flow MP4

Code	Type	Android-shim test point	Evidence (§11.4.5/.69/.107)
REC	recorded-evidence	driven by real TunnelStatus on device; consent-dialog flow window-scoped MP4 (§11.4.154/.155) of the real connect journey, StatusChip reading e.g. masque-h3 · 23 ms confirmed by vision/OCR (§11.4.163)	helixvpn---android-connect---<run-id>.mp4
CHAL/ HQA	Challenge / HelixQA (§11.4.27)	a challenges/helix_qa bank entry scoring PASS only on the captured on-device evidence above	bank result.json
DDOS / SCALE	—	NOT_APPLICABLE: single-node-selfhost (§11.4.3 topology-SKIP) — re-arm Phase 2	tracked SKIP-with-reason

Anti-bluff (§11.4.107/.158). A green Kotlin unit test is not proof the user is protected. The authoritative evidence is the **on-device** E2E+SEC+REC bundle: a window-scoped MP4 of the real connect journey on a real device, the pcap proving protect() keeps the carrier off the tun, and the curl-reaches-LAN capture — never a mock-claiming-PASS (O5, §11.4.27).

16. Open decisions & cross-doc seams

Per §11.4.6/§11.4.66 — surfaced with a recommendation, never silently resolved.

#	Decision	Option A	Option B
D- ANDROID-1	Process model	single process (core .so + VpnService + Flutter share one process)	:vpn separate service process
D- ANDROID-2	tun fd ownership	detachFd() → core owns + closes (§5.3)	getFd() → Kotlin owns

#	Decision	Option A	Option B
D- ANDROID-3	protect() mechanism	VpnService.protect upcall (typed, safe)	native protectFromVpn (perf)
D- ANDROID-4	Auto- reconnect on START_STICKY redelivery	always reconnect	reconnect only if last intent = connected (or lockdown)
D- ANDROID-5	Split-tunnel model	exclude-list (addDisallowedApplication, MVP)	include-list (addAllowedApplication, Phase 2)
D- ANDROID-6	API-34+ foreground- service type	(must be resolved before coding)	—

Cross-doc seams this shim fixes / consumes:

Seam	Direction	Counterpart
TunnelStatus (5-variant)	consumed, transparently forwarded	helix-core/src/ status.rs [01- ORCH §4.1] [03 §3.1]; the protect- callback is the Android- specific extension (§3.1, §6)
FFI surface (start/stop/ attach_tun/detach_tun/ status_stream) + the new set_protect_callback	consumed + the protect addition proposed here	
TunnelPlatform channel contract (startTunnel/ stopTunnel/events,	implemented	[03 §4]

Seam	Direction	Counterpart
TunnelConfig, PlatformTunnelEvent)		
RoamDetected{new_local}	produced by the shim's ConnectivityManager callback	consumed by the orchestrator [01-ORCH §7.5, §9]
device.revoked → OS- tunnel teardown	the OS-level onRevoke complements the core's Down{auth- failed}	doc 02 control plane + [01- ORCH §5.4]

Sources verified

- final/03-client-core-and-ui.md [03] — §0.1 (client invariants CI1-CI6), §3 (FFI surface, ClientConfig, attach_tun/detach_tun, the Dart TunnelStatus mirror + status-enum note), §4 (the TunnelPlatform contract, TunnelConfig, PlatformTunnelEvent, obligations O1-O5), §5.2 (the Android VpnService+JNI Kotlin sketch this doc deepens), §8 (Riverpod wiring), §10 (budgets), §11 (build/CI), §14 (D-CLIENT-3 binding generator).
- final/v02-data-plane/orchestrator-and-state.md [01-ORCH] — §4.1 (the frozen 5-variant TunnelStatus), §4.4 (Down.reason vocabulary), §4.6 (receiver contract / Lagged), §5.4 (Connected only after WG handshake; auth-failed), §7.5 (RoamDetected, < 3 s roam), §8 (kill-switch/DNS state machine), §2.4 (shutdown).
- final/v02-data-plane/transport-trait.md [01-TT] — §4.8 (effective_mtu per carrier → MTU 1280/1420), §6 (dial opens the socket the shim must protect), §7 (error taxonomy → ladder).
- v09-research/research-ios_android.md [research-ios_android] (all sources accessed 2026-06-25) — §3 (Android VpnService.Builder addAddress/addRoute/addDnsServer/setMtu/establish()→ParcelFileDescriptor; Layer-3 fd I/O; detachFd() vs getFd() ownership; protect() mandatory before connect; native protectFromVpn/NetdClient; Android has no iOS-NE-style jetsam cap), §4 (cross-platform synthesis; bound buffers to MTU+headroom; iOS is the binding constraint), §2 (size-first Rust profile shared with Android).
- 04_VPN_CLD/HelixVPN-Architecture-Refined.md [04_ARCH] — §5.3 shim matrix (Android = VpnService+JNI, Kotlin, ParcelFileDescriptor fd ⇌ core), §5.6 (kill-switch/DNS driven by core state machine; tunnel DNS forces, blocks off-tunnel :53).
- 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md [04_UI] — §6 shim row (Android core via JNI; foreground service), build/CI matrix.

- final/06-phase0-spike-wbs.md [06-WBS §0.4] + final/10-testing-acceptance-and-qa.md [10-QA §2] — the §11.4.169 closed test-type code map.
- v09-research/_SYNTHESIS.md [SYN] — §5 (client shim architecture), §9 (constitution bindings: §11.4.156 no-CI, §11.4.151/.155 prefixes, §11.4.10).

Constitution: §11.4.6 (no-guessing — every UNVERIFIED marked, none fabricated), §11.4.66 (decisions = options + recommendation), §11.4.10 (secret token handling), §11.4.14 (cleanup on every exit path), §11.4.99 (verify API-34 foreground type against latest docs), §11.4.107/.158/.159 (captured on-device window-scoped evidence), §11.4.169 (per-item test-type coverage), §11.4.156 (no active CI).

End of Volume 4 nano-detail specification — Android shim (VpnService + JNI). Pairs with [03 §5.2] (the overview this deepens), [01-ORCH §4] (TunnelStatus), and the sibling per-platform shim specs (apple/windows/linux/harmonyos/aurora). Surfaced decisions D-ANDROID-1...6 — presented, none silently resolved (§11.4.66).